Design and Implementation of an ESP32-Based Intelligent Knee Exoskeleton with BLE-Enabled Progressive Web App Controller

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***Abstract*—**This paper presents the design and implementation of a low-cost, ESP32-based intelligent knee exoskeleton system for lower-limb rehabilitation and mobility assistance. The system integrates an MPU6050 six-axis Inertial Measurement Unit (IMU) with a complementary filter for real-time limb pitch estimation and a servo motor for knee-joint actuation. A three-state finite state machine (FSM) — IDLE, WALKING, and SITTING — with hysteresis guards provides autonomous, context-aware assistive control. In Manual Mode, a Bluetooth Low Energy (BLE) Nordic UART Service (NUS) interface connects the device to a Progressive Web App (PWA) smartphone controller supporting direct angle control, servo speed adjustment, voice commands in four Indian languages, WebAuthn biometric authentication, emergency SOS geolocation, and live pitch telemetry charts. An SSD1306 OLED display provides on-device diagnostics. Battery health is monitored via a calibrated ADC voltage divider. Experimental results confirm reliable BLE connectivity (<30 ms latency), accurate pitch estimation (±1.8° RMS error), correct FSM state transitions under all simulated gait profiles, and smooth servo actuation — validating the platform as an effective, affordable solution for rehabilitation engineering research.

***Index Terms*—**Exoskeleton, ESP32, BLE, MPU6050, Complementary Filter, PWA, Web Bluetooth, Servo Actuator, Rehabilitation Robotics, Wearable Computing.

I. INTRODUCTION

Mobility impairment due to stroke, spinal cord injury, or age-related musculoskeletal degeneration affects an estimated 75 million people globally who require assistive walking devices [1]. Powered knee exoskeletons augment or replace muscle effort during rehabilitation and daily activity, accelerating recovery and improving independence. Commercial systems such as ReWalk, Ekso GT, and Cyberdyne HAL achieve impressive clinical outcomes but cost between $40,000 and $150,000 USD, confining their use to well-funded rehabilitation centres [1][4][6].

Recent convergence of low-cost microcontrollers, MEMS IMUs, Bluetooth Low Energy, and browser-native APIs (Web Bluetooth, WebAuthn, Geolocation) creates an opportunity to engineer effective, affordable alternatives accessible to community clinics and individual users. This work delivers a complete, end-to-end exoskeleton platform with the following contributions:

1) A self-contained ESP32 knee exoskeleton with MPU6050 IMU, servo actuation, OLED display, LED status indicators, and ADC battery monitor;
2) A dual-mode (Auto/Manual) firmware featuring complementary-filter IMU fusion, a hysteresis-guarded three-state FSM, non-blocking servo interpolation, and NVS-persistent settings;

D. Non-Blocking Servo Interpolator

The servo target is approached each loop iteration through a time-based degree-per-second interpolator, preventing blocking delays from stalling BLE or IMU tasks:

$$step_{max} = (1000 / delay_{ms}) \times \Delta t$$

The servo hardware register is written only when the rounded integer degree changes, minimising PWM jitter. Speed (0–100 ms/°) is BLE-configurable and NVS-persistent.

E. BLE Nordic UART Service Interface

The device advertises a standard NUS profile (Service UUID: 6E400001). Telemetry is notified on the TX characteristic at up to 2.5 Hz in Auto mode and event-driven in Manual mode:

ANGLE:<°>, BAT:<%>, VOLT:<V>, MODE:<X>, PITCH:<°>

TABLE III
BLE RX COMMAND SET

Command	Action
1 / 0	Assist (90°) / Relax (0°)
+ / −	Increment / decrement ±10°
A<n>	Set exact angle 0–180°
V<n>	Set servo speed (0–100 ms/°)
T<n>	Set tilt threshold (5–80°)
AUTO / MANUAL	Switch firmware mode
STAND / SIT / RELAX	Voice preset positions

BLE advertising is mode-gated: active only in MANUAL_MODE. On AUTO_MODE switch, the radio

3) A Progressive Web App smartphone controller with Web Bluetooth NUS, multi-language voice control, WebAuthn biometric lock, emergency SOS, live telemetry charts, and exportable activity logs;

4) Experimental validation across all sub-systems confirming clinical-grade responsiveness at a hardware cost under ₹2,000 (~$24 USD).

II. RELATED WORK

A. Powered Knee Exoskeletons

Electromechanical exoskeletons reviewed in [4] and [6] rely on dedicated embedded computers or tethered laptops for control. Recent academic prototypes using low-cost microcontrollers have demonstrated basic gait assistance [5], but few integrate smartphone wireless interfaces. Low-cost ESP32-based platforms [2][3] have shown promise though most lack a complete PWA controller, multi-language voice control, or biometric security as presented here.

B. IMU-Based Gait Detection

Gait phase detection using MEMS IMUs has been extensively studied [7][8]. Complementary filters, which blend high-frequency gyroscope integration with low-frequency accelerometer correction, are preferred on resource-constrained hardware over Kalman filters due to their substantially lower computational load while maintaining adequate angle accuracy for wearable applications [7][9]. Real-time lower-body inertial kinematics estimation has been validated with RMS errors below 2° under dynamic conditions [8].

C. BLE and Web Bluetooth for Wearables

Bluetooth Low Energy wearable systems have demonstrated sub-50 ms latency and reliable connectivity for real-time rehabilitation monitoring [10]. The Web Bluetooth API (W3C, 2023) exposes BLE GATT operations to JavaScript, enabling PWAs to connect to peripherals without native app development or app-store distribution [11]. Progressive Web Apps have been validated as a viable interface paradigm for mobile health applications, offering offline capability, home-screen installation, and access to native device APIs such as Geolocation and WebAuthn [12].

III. SYSTEM HARDWARE ARCHITECTURE

A. Central Processing Unit — ESP32

The ESP32 (Xtensa dual-core LX6, 240 MHz, 520 KB SRAM, 4 MB flash) integrates BLE 4.2, Wi-Fi 802.11 b/g/n, hardware PWM, I²C master, 12-bit ADC, and NVS flash storage — making it well-suited for simultaneous IMU acquisition, servo actuation, BLE communication, and display management within a single low-cost SoC.

B. MPU6050 IMU

The MPU6050 six-axis IMU (3-axis accelerometer ±2 g, 3-axis gyroscope ±250 °/s) communicates via I²C. The device address (0x68 or 0x69) is auto-detected during a boot-time bus scan. Registers 0x3B–0x48 (14 bytes: AcX, AcY, AcZ, Temp, GyX, GyY, GyZ) are read at 100 Hz. Power register 0x6B is written 0x00 to wake the device. A critical-error halt with continuous LED/buzzer is stopped and any active connection is forcibly terminated to conserve power.

V. PROGRESSIVE WEB APP CONTROLLER

A. Architecture and Connectivity

The controller is a single-page PWA (HTML5/CSS3/JavaScript) served over HTTPS and installable to Android home screens via *manifest.json* and a Cache API service worker for offline operation. Web Bluetooth scans by device name ("Exoskeleton") and subscribes to NUS TX notifications. An automatic 3-second reconnect loop recovers from unexpected disconnections without user intervention.

B. Real-Time Telemetry and UI

Incoming telemetry updates: (1) a colour-coded battery bar (green >40%, orange 16–40%, red ≤15%) with an audible voice warning at ≤15%; (2) an animated SVG semicircular arc (0°–180°) with a glowing dot tracking servo angle; (3) a 60-point rolling Chart.js time-series of live IMU pitch; (4) a mode badge that auto-disables manual control buttons when AUTO mode is reported by the firmware.

C. Voice Control and Multilingual Support

Continuous voice recognition uses the Web SpeechRecognition API. A critical conflict between Speech Synthesis (TTS feedback) and the microphone is resolved by pausing recognition during TTS playback and resuming after the utterance ends, preventing mic crashes. The Screen Wake Lock API prevents display sleep during active voice sessions. Four locales are supported: English (en-IN), Hindi (hi-IN), Kannada (kn-IN), and Telugu (te-IN).

D. Security — WebAuthn Biometric Lock

A lock screen on first load provides two authentication paths: (1) a 4+ character password stored in *localStorage*; (2) WebAuthn FIDO2 platform authenticator (fingerprint/face ID) via *navigator.credentials.create()* for registration and *navigator.credentials.get()* for login. The private key never leaves the device's secure enclave. Session state in *sessionStorage* keeps the app unlocked across page refreshes.

E. Emergency SOS

A floating SOS button captures GPS coordinates (Geolocation API, high-accuracy mode) and opens a pre-composed WhatsApp message to a saved emergency contact including a Google Maps link. A graceful fallback sends the message without coordinates if GPS is unavailable within the timeout.

VI. EXPERIMENTAL RESULTS

The complete system was tested on a bench-top rig simulating gait angles with a manually rotated IMU mount. Table IV summarises all key performance metrics.

TABLE IV
SYSTEM PERFORMANCE SUMMARY

Metric	Result
IMU pitch RMS error (static)	±1.8°
FSM state transitions verified	5 / 5 correct

alarm is triggered if the IMU fails to respond during boot.

C. Servo Actuator and OLED Display

A standard PWM servo motor (50 Hz, 500–2400 µs pulse) connected to GPIO 18 actuates the knee joint across 0°–180°. Four hardware PWM timers are allocated via the ESP32Servo library for precise pulse generation. An SSD1306 128×64 OLED (I²C) renders a 10 Hz dashboard displaying mode, limb pitch (°), servo target angle (°), activity state, and a battery charge bar. Table I provides the complete GPIO allocation.

TABLE I
GPIO PIN ALLOCATION

Component	GPIO	Function
Servo Motor	18	PWM actuation (0°–180°)
MPU6050 / OLED	21 / 22	I²C SDA / SCL
Mode Button	4 (PULLUP)	Short / long press
Alarm LED + Buzzer	19	Battery alert / beeps
Green LED	26	AUTO mode indicator
Blue LED	25	MANUAL mode indicator
Battery ADC	34	Voltage divider input

D. Battery Monitoring Circuit

A two-resistor voltage divider on GPIO 34 (ADC1_CH6, 12-bit) scales the 7.4 V LiPo pack voltage into the ESP32's 0–3.3 V ADC range. A software calibration factor (×3.0) corrects for resistor tolerances, confirmed by ADC measurement against a reference multimeter:

$$V_{bat} = (ADC_{raw} / 4095) \times 3.3 \times 3.0$$

Battery percentage is linearly mapped between full charge (8.4 V = 100%) and cutoff (6.0 V = 0%). A non-blocking alarm (1 Hz LED blink + chirp) activates below 6.5 V. A hard-floor guard (below 3.0 V) suppresses false alarms from a disconnected battery.

IV. FIRMWARE ARCHITECTURE

A. Boot Sequence and Dual-Mode Operation

On power-up, the firmware: (1) loads persistent settings (mode, servo speed) from ESP32 NVS flash; (2) configures GPIO and performs a 3-blink hardware self-test; (3) initialises the servo, I²C bus, MPU6050, OLED, and BLE stack; (4) enables BLE advertising only if the loaded mode is MANUAL_MODE. Mode is toggled by a long button press (≥2 s, 50 ms debounce) or a BLE voice command, and immediately written back to NVS to survive reboots.

B. Complementary Filter for Pitch Estimation

Limb pitch angle θ is estimated at 100 Hz by fusing accelerometer and gyroscope data using a complementary filter (α = 0.96). Raw 16-bit registers are converted to physical units using the default full-scale ranges (±2 g → ÷ 16384, ±250 °/s → ÷ 131):

$$\theta_{acc} = atan2(A_x, \sqrt{(A_y^2 + A_z^2)}) \times 180/\pi$$

$$\omega = G_y / 131.0 \ [°/s]$$

$$\theta_k = \alpha \cdot (\theta_{k-1} + \omega \cdot \Delta t) + (1-\alpha) \cdot \theta_{acc}$$

High-precision timing uses *micros()* with Δt clamped to (0, 100 ms] to handle scheduling jitter safely. The 96% gyroscope weighting captures fast dynamic limb motion, while the 4% accelerometer term continuously corrects long-term gyroscope bias drift.

SITTING false positives (walking)	0 / 20 trials
BLE indoor range (LOS)	≥ 8 m
BLE command latency	< 30 ms
90° servo sweep (10 ms/°)	≈ 900 ms
StandBoost sweep (2 ms/°)	≈ 180 ms
Battery measurement error	±0.15 V (±7%)
Voice command accuracy	> 90% (quiet room)
BLE auto-reconnect time	3 – 5 s

The 2-second SITTING debounce timer eliminated all false positives across 20 simulated walking trials. The hysteresis band (70°–75°) successfully maintained SITTING state during minor postural oscillations. StandBoost provided a measurable assistive force impulse during the sit-to-stand transition, reducing perceived effort in informal user feedback.

VII. DISCUSSION

A. Key Strengths

The platform achieves a complete closed-loop assistive system — from IMU sensing and real-time fusion to autonomous actuation and smartphone telemetry — at a total hardware cost under ₹2,000 (~$24 USD). The PWA approach eliminates app-store distribution overhead while providing access to native APIs. Mode-gated BLE reduces radio-frequency interference and power consumption during autonomous operation.

B. Limitations and Future Work

Current limitations include: (1) single-axis sagittal-plane control only; (2) hobby-grade servo torque insufficient for high-resistance rehabilitation; (3) complementary filter drift under sustained vibration. Future work will integrate FSR insole sensors for ground-contact gait phase detection, TensorFlow Lite Micro for on-device gait classification, cloud telemetry over Wi-Fi for caregiver remote monitoring, and a torque-feedback brushless motor for clinical-grade actuation.

VIII. CONCLUSION

This paper presented a complete ESP32-based knee exoskeleton integrating complementary-filtered IMU sensing, a hysteresis-guarded three-state activity FSM, BLE NUS communication, and a PWA mobile controller with voice control, biometric authentication, live telemetry, and emergency SOS. All sub-systems were experimentally validated. The platform demonstrates that clinical-grade assistive responsiveness is achievable on commodity hardware costing under $25 USD, making it a strong open-source foundation for rehabilitation research and low-resource assistive technology deployment.

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The absolute pitch magnitude |θ| drives a three-state FSM. Hysteresis bands prevent oscillation at state boundaries, and a 2-second confirmation timer prevents false SITTING detection during transient bends common in walking:

TABLE II
FSM STATE TRANSITION RULES

State	Entry Condition	Servo Target
IDLE	$\|\theta\| < 8°$	0° (Relaxed)
WALKING	$8° \leq \|\theta\| < 75°$	map(|θ|, T_{up}, 90, 15, 130)
SITTING	$\|\theta\| \geq 75°$ for >2 s	above × 0.95 (anti-hum)

Target angle is linearly mapped and constrained to [0°, 180°]. A 1.5° dead-band on *targetServoAngle* suppresses noise-induced micro-movements. A **StandBoost** flag — activated on SITTING→WALKING — temporarily increases servo speed to 2 ms/° (vs. default 10 ms/°) and adds a +10° overshoot to actively assist the sit-to-stand transfer, cancelling automatically once |θ| drops below 35°.

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